

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Seventh Semester B.E. (Mechanical) (Credit System) Degree Examinations

January 2023

17ME702 – MECHANICAL VIBRATIONS

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **One full** question from **each Unit**.

2) Any missing data may be assumed suitably.

Unit – I

1. a) Define (i) Degrees of freedom (ii) Fundamental mode of vibration
- b) A body is subjected to 2 harmonic motions
 $x_1 = 15\sin(\omega t + \pi/6)$, $x_2 = 8 \cos(\omega t + \pi/6)$, what harmonic is to be given to the body to it to equilibrium.
- c) Find the natural frequency for the system shown in Fig. 1 (c).
 Use both Energy & Newton's method.

Marks	BT*	CO*	PO*
2	L*1	1	1

8	L2	1	1
---	----	---	---

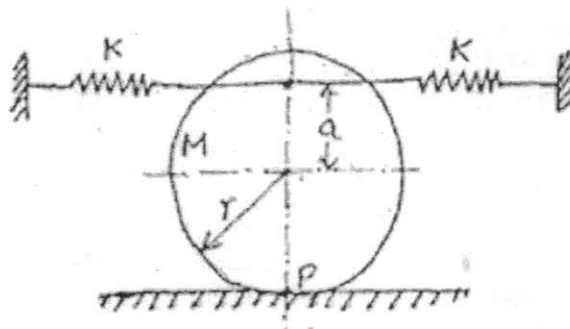


Fig. 1 (c)

10	L3	1	1
----	----	---	---

2. a) Define: i) Forced Vibrations ii) Resonance
- b) What's the effect of mass of a spring on its natural frequency? Derive the equation for natural frequency.
- c) Determine the natural frequency of the system shown in Fig. 2(c) by Newton's method and Energy method.

2	L1	1	1
---	----	---	---

8	L1	1	1
---	----	---	---

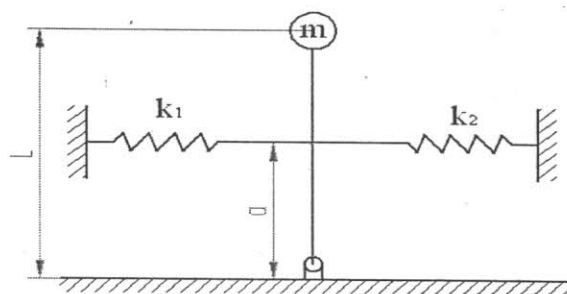


Fig. 2 (c)

10	L3	1	1
----	----	---	---

Unit – II

3. a) Obtain the equation of motion for an under damped system. Also obtain the complete solution.
- b) A vibrating system has following characteristics:
 $W = 19.62 \text{ N}$, $k = 5000 \text{ N/m}$, $C = 60 \text{ Ns/m}$.
 Determine: i. Damping factor
 ii. Damped natural frequency
 iii. Logarithmic decrement
 iv. Ratio of two successive amplitudes.

10	L1	2	1
----	----	---	---

10	L3	2	2
----	----	---	---

4. a) The measurement on a mechanical vibrating system shows that the mass of 10 kg and that the springs can be combined to give an equal spring stiffness of 5 N/mm. If the vibrating system have a dashpot attached which exerts a force of 40 N at a velocity of 1 m/s. Determine (i) critical damping coefficient (ii) damping factor (iii) Logarithmic decrement (iv) Ratio of any two consecutive amplitudes
- b) A torsional pendulum has a natural frequency of 200 cycles/min, when vibrating in vacuum. The mass moment of inertia of the disc is 0.2 kgm^2 . It is then immersed in oil and its natural frequency is found to be 180 cycles/min. Determine the damping constant. If the disc, when placed in oil is given an initial displacement of 2° , find the displacement at the end of the first cycle.

10	L3	2	2
10	L3	2	2

Unit – III

5. a) Derive an expression for determining the amplitude of vibration in a machine with rotating unbalance.
- b) A heavy machine of 3000 N, is supported on a resilient foundation. The static deflection of the foundation due to the weight of the machine is found to be 7.5 cm. It is observed that the machine vibrates with amplitude of 1cm when the base of the machine is subjected to harmonic oscillations at the undamped natural frequency of the system with amplitude of 0.25 cm. Find (a) the damping constant of the foundation (b) the dynamic force amplitude on the base (c) the amplitude of the displacement of the machine relative to the base.
6. a) Define displacement transmissibility. Deduce an expression for the same. Sketch its variations with frequency ratio for different amounts of damping.
- b) The support of a spring mass system is vibrating with an amplitude of 5 mm and a frequency of 1150 cpm. If the mass is 0.9 kg and the stiffness of springs is 1960 N/m, Determine the amplitude of vibration of mass. What amplitude will result if a damping factor of 0.2 is included in the system.

8	L2	3	1
12	L3	3	2
10	L2	3	1
10	L3	3	2

Unit – IV

7. a) Derive the expression for critical speed of a shaft considering damping effect.
- b) Mention the instruments used to measure displacement, velocity and acceleration. List the needs of vibration measuring instruments.
8. a) Set up the differential equation of motion for the system shown in Fig. 8(a) and hence derive the frequency equation & obtain the two natural frequencies. Assume, $m_1 = m_2 = m \text{ kg}$ & $l_1 = l_2 = 1 \text{ m}$.

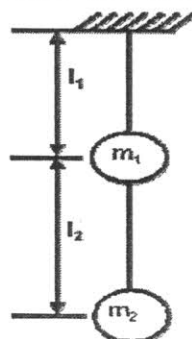


Fig. 8(a)

15	L3	4	2
----	----	---	---

- c) Write a note on undamped vibration absorber.

5 L1 4 1

Unit – V

9. a) Find first natural frequency and modal vector of the system shown in the Fig. 9(a) using matrix iteration method. Use flexibility influence coefficients.

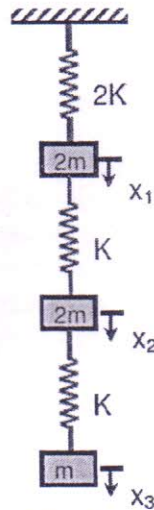


Fig. 9 (a)

15 L3 5 2

- b) Explain the steps in Stodola method to obtain fundamental natural frequency.
10. For the system shown in the Fig.10, obtain natural frequencies using Holzer's method. Take $J_1 = J_2 = J_3 = 1 \text{ kgm}^2$ and $k_1 = k_2 = 1 \text{ Nm/rad}$, Assume trial frequency as 0.25 rad/s .

5 L1 5 1

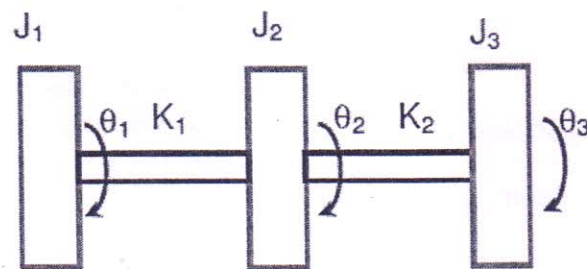


Fig. 10.

20 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
